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An integrated group decision-making process for supplier selection and order allocation using multi-attribute utility theory and linear programming

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Abstract

Supplier selection is a complex multi-criteria problem including both quantitative and qualitative factors. In order to select the best suppliers it is necessary to make a trade-off between these factors, some of which may be in conflict and may also be uncertain. In this problem if suppliers have capacity or other different constraints, two problems will exist: which suppliers are best and how much should be purchased from each selected supplier.

In this paper an integrated approach of multi-attribute utility theory (MAUT) and linear programming (LP) is proposed for rating and choosing the best suppliers and defining the optimum order quantities among selected ones in order to maximize total additive utility. A numerical example is proposed to illustrate an application of the proposed method.
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1. Introduction

In today's fierce competitive environment, characterized by thin profit margins, high consumer expectations for quality products and short lead-times, companies are forced to take advantage of any opportunity to optimize their business processes. To reach this aim,

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academics and practitioners have come to the same conclusion: for a company to remain competitive, it has to work with its supply chain partners and improve the chain's total performance. When the main process in the upstream chain that affects all parts of an organization is considered, the purchasing function emerges as increasingly important. Thus, supply chain management and the supplier (vendor) selection process have received considerable attention in the business-management literature.

Supplier selection is a fundamental issue in the supply chain which heavily contributes to the overall supply chain performance. Companies who spend a high percentage of their sales revenue on parts and material supplies, and whose material costs represent a larger portion of total costs, savings from supplies are of particular importance. These situations require more systematic and transparent approach to purchase decision making, especially regarding the area of supplier selection. Carefully selecting the suppliers significantly reduces the purchasing cost and improves corporate competitiveness which is why many experts believe that supplier selection is the most important activity of a purchasing department. Supplier selection is the process by which suppliers are reviewed, evaluated and chosen to become part of the company's supply chain.

Several factors affect a supplier's performance. The analysis of such criteria and supplier performance measurement has been the focus of many scientists and purchasing practitioners since the 1960s. As an important work, Dickson [1] presented 23 supplier selection metrics. Dickson's paper has been used as a reference for the majority of the papers dealing with the supplier selection problem. Years afterward, Roa and Kiser [2], Ellram [3], Stamm and Golhar [4] identified 60, 18, and 13 criteria for supplier selection, respectively. However, supplier selection is risky and complicated due to uncertainty and conflict between several quantitative and qualitative factors.

Organizations have two approaches to supplier selection. The first approach is to select the best single supplier, which can meet all the requirements (single sourcing). The second approach is to select an appropriate combination of suppliers when no single supplier can satisfy all the requirements. Accordingly, management should split order quantities among the available suppliers for a variety of reasons including creating a constant environment of competitiveness (multiple sourcing).

Several methods have been proposed in the literature for single sourcing supplier selection some of which are discussed. One of these methods is data envelopment analysis (DEA), DEA is a mathematical programming (MP) technique that calculates the relative efficiencies (ration of weighted outputs (benefit criteria) to weighted inputs (cost criteria)) of multiple decision-making units. Weber and Ellram [5] primarily discussed DEA usage in supplier selection. Also Liu et al. [6], Forker and Mendez [7] and Saen et al. [8] used DEA for rating and choosing the best supplier. Talluri and Narasimhan [9] stated that methods such as DEA have primarily relied on evaluating vendors based on their strengths and failed to incorporate their weaknesses into the selection process. They also added that such approaches would not be able to effectively differentiate between vendors with comparable strengths but significantly different weaknesses.

The analytical hierarchy process (AHP) introduced by Saaty [10], based on pairwise comparison has been applied to supplier selection. Narasimhan [11], Partovi and Banerjee [12], Nydick and Hill [13], Barbarosoglu and Yazac [14], Yahya and Kingsman [15], Masella and Rangoue [16], Tam and Tummda [17], and Lee et al. [18] proposed to use this technique to cope with determining scores. Also, Liu and Hai [19] presented the voting analytical hierarchy process (VAHP), is a novel easier weighing procedure in place of

AHPs paired comparison. The analytical network process (ANP), a more sophisticated version of AHP introduced by Saaty (1996) [39], was applied for vendor selection by Sarkis and Talluri [20]. Despite the numerous advantages of these approaches, they have some short-comings. For example they don't effectively take into account risk and uncertainty when assessing the supplier's potential performance because they presume that the relative importance of attributes affecting the supplier's performance is known with certainty [21].

Multi-attribute utility theory (MAUT) has been used in international supplier selection considering one decision maker in a single sourcing environment by Min [21] however the probabilistic nature of supplier performance still remained unanswered.

In multiple sourcing, some authors have applied different methods of MP such as linear programming (LP) [22,23], mixed integer LP [24], multi-objective programming (MOP) [24,25], and goal programming (GP) [26,27]. One of the disadvantages of these methods is their failure to consider qualitative factors, making it unclear for managers to analyze them [28].

Hybrid models have also been applied to deal with this problem. Ghodsypour and O'Brien [23] combined AHP and LP in order to take into account tangible and intangible criteria and to optimize order allocation among suppliers. In another article, they proposed a hybrid model based on AHP combined with mixed integer nonlinear programming [24]. Other hybrid techniques have been used for solving supplier selection in multiple sourcing environments and order allocation such as DEA and MOP [29], AHP and GP [27], and AHP and mixed integer MOP [30]. Usually the disadvantages of combined methods affect the effectiveness of hybrid models.

Although many models have been proposed to deal with supplier selection problem, the probabilistic nature of the problem, qualitative and quantitative factors and uncertainty involved with it are not considered simultaneously in a multiple sourcing environment. In this paper, an integrated MAUT approach and a LP model are proposed for rating and choosing the best suppliers and indentifying the optimum order quantities among the selected suppliers in order to maximize total additive utility. Furthermore, not only are both qualitative and quantitative factors under risk and uncertainty considered, but the probabilistic nature of supplier performance is taken into account.

The remaining sections of the paper are structured as follows: In the section 2, an overview and background of MAUT method is presented. Section 3 will focus on the proposed model. In Section 4, a numerical example is illustrated. Then results are discussed in Section 5. In the final section, some conclusions are drawn from the study.

2. Multi-attribute utility theory (MAUT)

As indicated in the previous section, supplier selection is a complex decision-making problem. The complexity stems from the probability nature of the problem, a multitude of quantitative and qualitative factors influencing supplier choices as well as the intrinsic difficulty of making numerous tradeoffs among these factors. One analytical approach often suggested for solving such complex problems is MAUT.

Keeney and Raiffa [31] employed the utility concept in complex decision problems involving multiple attributes and multiple conflicting objectives, and provided a systematical approach of multiple attributes utility analysis (MAUA). MAUA is targeted in solving problems of trading off the achievement of some objectives against other objectives to obtain the maximum overall utility. A decision-maker is assumed to be facing

the above-mentioned problem, and he/she has to choose a solution from some solution alternatives. MAUA is used to assess the decision-maker's preference structure and model it mathematically with a multiple attributes utility function. This multiple attributes utility function is then applied to help the decision maker reach an optimal decision.

MAUT enables the decision maker to structure a complex problem in the form of a simple hierarchy and to subjectively evaluate a large number of quantitative and qualitative factors in the presence of uncertainty. Utility is a measure of desirability or satisfaction and provides a uniform scale to compare and/or combine quantitative and qualitative factors. A utility function is a device which quantifies the preferences of a decision maker by assigning a numerical index to varying levels of satisfaction of a criterion [32].

The major strength of MAUT is its ability to deal with both deterministic and stochastic decision environments [33]. In particular, Min [21] highlighted three distinctive advantages of MAUT over MOP in handling multiple and conflicting criteria as follows:

(1) MAUT requires less "front-end" analysis than MOP, as MAUT has no constraints considering explicitly;
(2) MAUT requires data more than MOP, as MAUT does not necessitate parameters for constraints; and
(3) MAUT poses less computational difficulty than MOP, as MAUT is not burdened with additional constraints.

The systematic nature of MAUT in tackling complex problems with conflicting criteria makes it especially suitable for selecting the most appropriate supplier.

3. An integrated method for supplier selection and order allocation

The model presented in this section applies MAUT, which unifies quantitative and qualitative factors using the same scale to measure desirability of the suppliers. The model is intended to rate the suppliers while considering uncertainty. The obtained rates are then used as coefficients for the objective function of the proposed LP model. The solution obtained from the maximum total additive utility (TAU) of the LP model provides the optimal allocation of the order quantities among the suppliers. The process is depicted in Fig. 1, and the main steps of the algorithms are described in the following sections.

3.1. Identify the problem scope and define the objectives or goals

In this step, the scope of the problem is defined in terms of the product/service to be outsourced, time frame for outsourcing, justification of decision, constraints in the supplier selection process, and available alternative sources to choose from. Then the objectives of supplier selection are derived from the various functional departments of the organization, which are aligned with the overall organizational goals and impacted by the decision: such as the assembly line, supplier quality assurance department, finance group, logistics department and the other related sections.

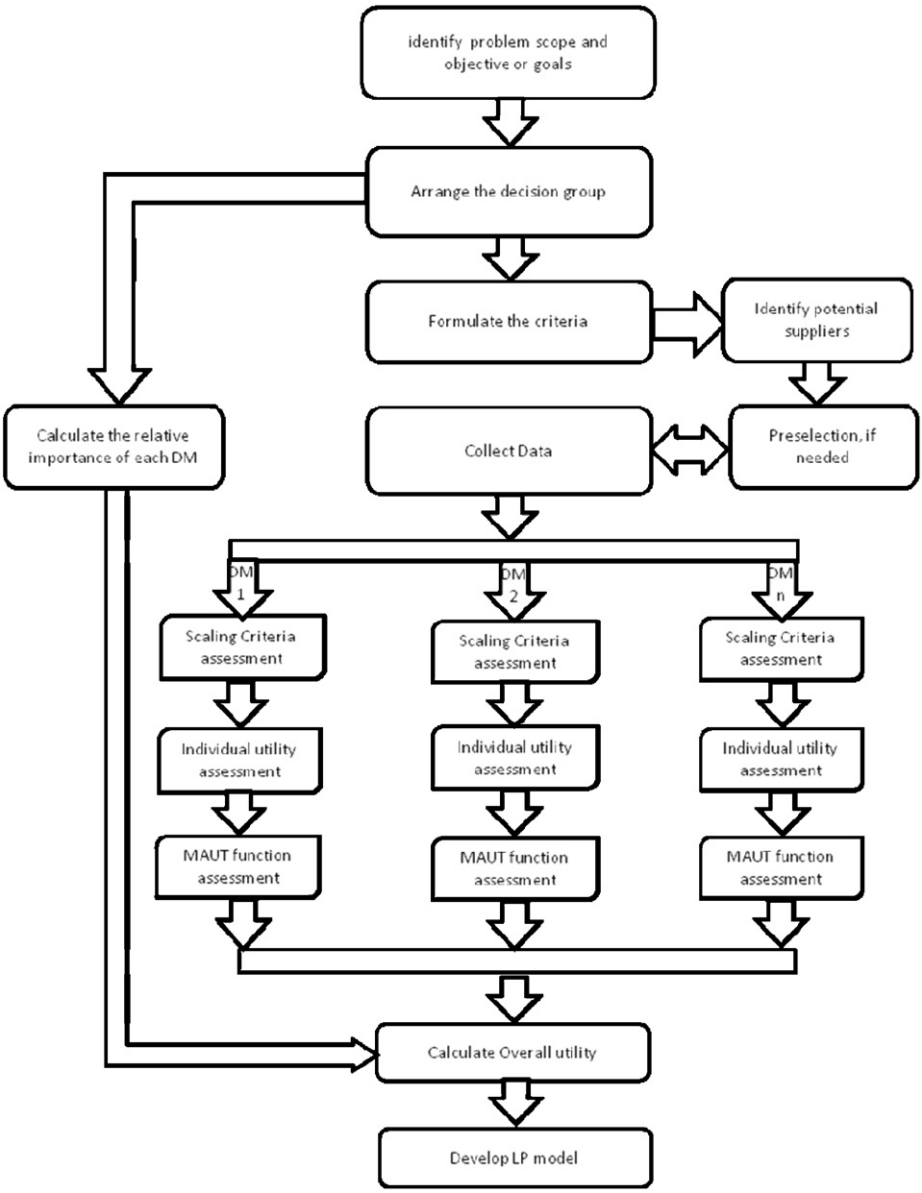

Fig. 1. An integrated method for supplier selection and order allocation.

3.2. Define and describe a finite set of relevant attributes

For supplier selection, identification of decision attributes (criteria) is required, and then evaluation scales/metrics are determined in order to measure appositeness of a supplier. These criteria must be defined according to the corporate strategies and the company's

competitive situation. In this way, the level of buyer–supplier integration [23] is determined and the product type outsourced. Having considered sub-criteria for each main criterion, a hierarchical form called "value tree" is structured along with a unit to measure each criterion. The objectives and goals identified in step 3.1 and the decision makers' desires have direct impacts on the degrees to which these attributes are encountered.

3.3. Pre-selection of potential suppliers

According to Aissaoui et al. [34] today's co-operative logistics environment requires a low number of suppliers as it is very difficult to manage high numbers. Therefore, the purpose of this stage of the process is to rule out the inefficient candidates and reduce the set of all suppliers to a small range of acceptable ones.

Among the existing supplier alternatives, it is possible to use an elimination method which excludes suppliers who cannot satisfy the selection rules. The concept is based on the fact that suppliers who do not fit the minimum level of key criterion/criteria are gradually eliminated during the selection process. From a practical viewpoint, the approach reduces the number of considerations and helps to focus on highly competitive options. To achieve this aim a brief data collection with respect to key criterion/criteria is required. De Boer et al. [37] and Aissaoui et al. [34] discussed the methods supporting pre-selection of potential suppliers.

3.4. Data collection

The data are collected from each qualified supplier through interviews, questionnaires, personal meetings, visits to the supplier facility and external resources. The best/worst levels rather than maximum/minimum levels should be identified in this step. The extrapolation of the utility outside the measurement boundaries cannot be reliably accomplished. The data collection could also facilitate the estimate of uncertainty concerning the supplier performance.

3.5. Utility assessment

The utility assessment step is the decisive part of the supplier selection process. This step is conducted through a series of interviews with the decision makers by an expert analyst. The parameters of the utility function are calculated directly from a lottery and trade-off questions. Depending on the nature of individual attributes and the assumptions regarding preferential and utility dependencies, a multi-attribute utility function can be derived.

The MAUA method is based on a two-step approach of mapping the alternatives to the overall utility values as follows:

i. Establish a utility function for each attribute by associating probabilities with the attribute levels, or a value function by a direct assessment method. The value of each attribute's performance with respect to the decision maker is elicited in terms of preference/scaling factors. The utility values of an individual attribute for each alternative supplier are assessed through an interview process.

ii. Find an appropriate form to combine single attribute utility/value functions into an overall utility value.

For details on the application of multi-attribute utility function forms and consistency checks, please refer to Keeney and Raiffa [31]. The derived multi-attribute utility function is utilized to calculate the final utility score for each supplier. This utility score is used to rate the suppliers.

3.6. Compute the overall (surrogate) utility of group decision making

Multiple decision makers are often preferred to a single decision maker. To calculate the overall utility when more than one decision maker is involved with the supplier selection process, as is often the case, utility scores elicited from each decision maker should be considered. Surrogate utility function (SUF) for group decisions has the weighted additive form as follows [35]:

$$U(x) = \sum_{i=1}^{K} a_i u_i(x) \tag{1}$$

where i is the decision maker's index, K the number of decision makers, u_i (x) the utility function of ith DM, and a_i the weight of ith DM.

To use this function, it is necessary to set values for a_i s, the weights assigned to individual utility functions. Bodily [35] noted that these weights may be assigned either by a super decision maker (benevolent dictator) or through mutual agreement of the committee members. He proposes a delegation process to find a_i s by solving the system of equations below:

$$a = a\,\underline{W} \quad \textit{and} \quad \sum_{i=1}^{K} a_i = 1 \tag{2}$$

where W is a delegation matrix in which each committee member within a subcommittee designates voting weights to the other members. Individual i assigns weight w_{ij} to member j of his delegation subcommittee, when $0 \leqslant w_{ij} \leqslant 1$, $j = 1,2,\ldots,K$; $w_{ii} = 0$; $\sum_{j=1}^{K} w_{ij} = 1$; for all i; and $\mathrm{a} = [a_1, a_2, \ldots, a_{\mathrm{K}}]$.

3.7. Build a LP model to maximize TAU

An LP model is formulated to assign order quantities to the suppliers. If there are some constraints such as suppliers' capacity, quality, etc., use the suppliers' ratings as coefficients of the LP objective function in order to assign optimal order quantities to the suppliers. As a result, the TAU becomes maximized while purchasing as much as we can from the most desirable suppliers. The objective function and constraints of this LP model are as follows:

3.7.1. Notations

U_i final utility (rate) of ith supplier
X_i order quantity for ith supplier
V_i capacity of ith supplier

C_i	total logistic cost of ith supplier
D	demand for the period
q_i	defect percent of ith supplier
Q	buyer's maximum acceptable defect rate
B	total budget

3.7.2. Objective function

As U_i and X_i, respectively, denote the ratings for and the number of purchased units from the ith supplier and maximizing the TAU as the objective function of desired purchasing as follows:

$$\text{Max(TAU)} = \sum_{i=1}^{n} U_i X_i \tag{3}$$

3.7.3. Constraints

The important constraints of the problem are supplier capacity, buyer's demand and quality [23], which are formulated as follows:

3.7.4. Capacity constraint

As vendor i can provide up to V_i units of the product, the order quantity (X_i) should be equal or less than the vendor capacity as follows:

$$X_i \leqslant V_i; \quad i = 1, 2, \ldots, n. \tag{4}$$

On the other hand, the aggregate suppliers' capacity should be equal or greater than the demand. Therefore:

$$\sum_{i=1}^{n} V_i \geqslant D \tag{5}$$

3.7.5. Demand constraint

The sum of the assigned order quantities to n vendors should meet the buyer's demand as stated below:

$$\sum_{i=1}^{n} X_i = D \tag{6}$$

3.7.6. Quality constraint

Since Q is the buyer's maximum acceptable defect rate and q_i is the defect rate of the ith vendor, the quality constraint can be expressed as follows:

$$\sum_{i=1}^{n} X_i q_i \leqslant QD \tag{7}$$

3.7.7. Cost constraints

Cost plays an essential factor in the supplier selection. Budget allocated for outsourcing or buying products is not unlimited.

Consider B as total allocated budget and C_i total logistic cost of ith supplier which involves products cost, ordering cost and shipping cost. Thus

$$\sum_{i=1}^{n} C_i X_i \leqslant B \tag{8}$$

3.7.8. Final model

The final integrated LP model can be shown as

$$\begin{aligned}
\text{Max(TAU)} = \sum_{i=1}^{n} U_i X_i \\
\text{Subject to:} \quad & \sum_{i=1}^{n} X_i = D \\
& \sum_{i=1}^{n} V_i \geqslant D \\
& X_i \leqslant V_i; \quad i = 1, 2, \ldots, n \\
& \sum_{i=1}^{n} C_i X_i \leqslant B \\
& \sum_{i=1}^{n} X_i q_i \leqslant QD \\
& X_i \geqslant 0; \quad i = 1, 2, \ldots, n
\end{aligned} \tag{9}$$

4. Numerical example

To illustrate the model implementation, the supplier selection approach proposed in this paper is used for a typical supplier selection problem. *XYZ* Inc. is an automobile manufacturer, which purchases parts and materials having a large portion of the total cost. For the company to survive in the competitive environment the savings from supplies is particularly vital.

To apply the methodology, a committee of three decision-makers from different functional departments of the company is formed. As the first step the actors must be aligned with the overall organizational goals. Having discussed the overall organization goal, a reasonable set of the comprehensive objectives for supplier selection process is formulated. These objectives must be in line with the overall organization goal. The next step involves specifying attributes to measure the degree to which these several objectives are met. The supply chain operations reference (SCOR) model can be used to identify these attributes. The SCOR model is a descriptive model constructed by the supply chain council (SCC) and intended to be an industrial standard [36]. The criteria, attributes and their definitions are listed in Table 1.

The attributes are categorized into five groups and described as follows:

Reliability The criterion regarding the performance of the supplier in delivering the correct product with the correct quantity, to the correct place, at the correct time, and in the correct condition and packaging.

Table 1
Attributes descriptions

Category	Attribute	Description	Symbol
Reliability	Damage free orders	Number of orders received damage free divided by total number of orders processed in the specific time period	X_1
	On time orders	Number of orders received on time divided by total number of orders in the specific time period	X_2
Responsiveness	Lead time	The average actual required time from the moment the supplier receives an order to the moment it ships	X_3
	Return product velocity	Average time required for process of returning the defective, incomplete or damaged orders and reshipping of the order to customer	X_4
Flexibility	Order increase/ decrease flexibility	Average time required to achieve an unplanned sustainable 30% increase/decrease in orders	X_5
	Revise flexibility	Ability to change or revise in the production operations	X_6
Cost/Financial	Total cost	Including component cost, shipment cost, order cost etc.	X_7
	Payment terms	Suitability of terms and conditions regarding payment of invoices, open accounts, sight drafts, credit letter and payment schedule	X_8
Infrastructure	Quality system certification	Quality certifications acquired	X_9
	Company size and reputation	Including good reputation, facility size, turnover, capacity etc.	X_{10}

Responsiveness The criterion related to the velocity at which a supplier provides products to the customer.

Flexibility The criterion regarding the agility of a supplier in responding to the demand changes.

Cost/Financial The criterion regarding cost and financial aspects of procuring from suppliers.

Assets/Infrastructure The criterion regarding the effectiveness of suppliers in managing assets to support the original equipment manufacturer (OEM) demand.

Having identified the attributes, the utility values should be assessed. The MAUT assessment is conducted through a series of interviews with three decision makers by an expert analyst. For the illustration purpose, a simple additive form of a multi-attribute utility function is employed. Additive utility theory (AUT) is chosen for the following two reasons:

(1) AUT provides a more practical methodology due to easier computational analysis.
(2) AUT is easier to understand and explain to decision makers.

The additive multi-attribute utility function is represented as

$$U_i = \sum_{j=1}^{M} (K_j(U_i(X_j)) \tag{10}$$

where i is the supplier index, j the attribute index, K_j the preference factor or weight of j th attribute, M the number of attributes, and U_i the overall supplier score.

The evaluation of a preference/scaling factor is an iterative process, which is described by Keeney and Raiffa [31]. A similar interview processes are used to understand the weight of each attribute for the decision maker. It is based on solving simultaneous equations derived from the fact that equally preferred options must have equal overall utility values. This is accomplished by asking the decision maker to compare a consequence of one attribute at the most preferred amount and all of the other attributes at the least preferred level to a lottery yielding having probability p and probability $1-p$, respectively. Probability p^* at which the decision maker is indifferent to this lottery is used for equating the overall utilities of two consequences to arrive at scaling factor for the attribute. This procedure also needs a number of consistency checks and iterations for refining the results which may make it time consuming and complicated. With n attributes, $n-1$ scaling constants need to be determined using the simultaneous equations. The last factor can be calculated from the constraint, i.e. the sum of all scaling factors equals 1. This analysis can be simplified by using software tools like Logical Decision. These kinds of decision support tools can navigate the decision maker through various steps with interactive graphical interface to arrive at final values of scaling factors [21].

By calculating the preference/scaling factor for each alternative supplier, the utility values for each supplier are assessed. Various techniques can be applied in determining these utility values [31]. In this case, the method of direct assessment is used. The utility values for the two extreme alternatives are assigned first, typically allocating 0 for the least preferable alternative, and 1 for the most preferable alternative. As a result, for each alternative, a probability value (p') is assessed when a decision maker is indifferent to lottery of choosing that particular alternative versus choosing most preferable alternative with probability Π or least preferable alternative with probability $1-p'$. This assessment is done through an interview process with the decision maker. Extracting exact preferences of the decision maker is very critical and needs expertise, aptitude and dedication on the part of analyst, as well as, decision maker. Due to space constraints only final results for the utility values of all suppliers derived from three decision makers with respect to ten attributes are shown in Table 3.

When considering the inherent uncertainty for the supplier performance, the following methodology is proposed. For uncertain attributes, decision makers may face different utility values for various performance levels. The past data is used to indicate those levels and the corresponding probability values as shown in the following formula:

$$U_i(X_j) = \sum_{l=1}^{L} (p_{ijl} U_i(X_{jl})) \tag{11}$$

where i is the supplier index, j the attribute index, l the level index in attribute j, L the number of levels in attribute j, $p_{ij\mathrm{n}}$ probability of level index in attribute j.

For example the lead time performance metric (X_2) represents the average actual order lead time. For supplier S1 there is a 50% chance of being less than 7 days, a 30% chance of

being between 7 days and 14 days, and a 20% chance of being more than 14 days. The DM1's utility values for various levels explained above are assessed via interviews and depicted in Table 2.

Utility value for supplier S_2 with respect to attribute X_2 would be:

$$(0.5 \times 0.8) + (0.3 \times 0.67) + (0.2 \times 0.3) = 0.66$$

Similar methodology is implemented for all the uncertain metrics (Table 3).

Table 4 shows the final utility values for each supplier. So far, the utility values for each supplier with respect to each attribute have been calculated. Consequently, the overall utility of the decision-making group must be computed to arrive at a single number representing the overall supplier score. This number will guide the decision by rating each supplier.

Consider that W is decision makers' delegation matrix.

$$W = \begin{pmatrix} 0 & 2/3 & 1/3 \\ 1/3 & 0 & 2/3 \\ 1/2 & 1/2 & 0 \end{pmatrix} \quad \text{So } a = [\,12/41 \quad 15/41 \quad 14/41\,]$$

Therefore, the final DMs importance weight will be:

$$(\,12/41 \quad 15/41 \quad 14/41\,)\begin{pmatrix} 0.496 & 0.503 & 0.404 \\ 0.455 & 0.477 & 0.299 \\ 0.396 & 0.398 & 0.279 \end{pmatrix} = (\,0.447 \quad 0.457 \quad 0.323\,)$$

Hence, the final utility for each supplier is calculated.

Table 2
DM1s utility values for each performance level of attribute X_2 with their related probability

Performance level	Probability	Utility
Less than 7 days	0.5	0.8
Between 7 days and 14 days	0.3	0.67
More than 14 days	0.2	0.3

Table 3
The utility values for the all suppliers derived from the three decision makers with respect to the ten attributes

Attribute	DM1				DM2				DM3			
	K_j	$U_1(X_j)$	$U_2(X_j)$	$U_3(X_j)$	K_j	$U_1(X_j)$	$U_2(X_j)$	$U_3(X_j)$	K_j	$U_1(X_j)$	$U_2(X_j)$	$U_3(X_j)$
X_1	0.090	0.840	0.730	0.349	0.110	0.950	0.870	0.238	0.060	0.750	0.680	0.345
X_2	0.090	0.660	0.726	0.540	0.074	0.720	0.630	0.490	0.090	0.600	0.530	0.390
X_3	0.027	0.700	0.870	0.389	0.046	0.670	0.770	0.263	0.048	0.540	0.628	0.163
X_4	0.107	0.250	0.612	0.420	0.106	0.150	0.523	0.320	0.072	0.300	0.580	0.480
X_5	0.065	0.547	0.940	0.234	0.124	0.355	0.543	0.138	0.070	0.687	0.894	0.197
X_6	0.098	0.456	0.796	0.547	0.067	0.398	0.659	0.412	0.070	0.287	0.894	0.327
X_7	0.338	0.542	0.187	0.462	0.248	0.483	0.143	0.399	0.390	0.334	0.060	0.253
X_8	0.085	0.274	0.367	0.203	0.106	0.189	0.434	0.115	0.130	0.256	0.382	0.195
X_9	0.050	0.326	0.528	0.197	0.054	0.426	0.497	0.156	0.025	0.386	0.532	0.174
X_{10}	0.050	0.246	0.454	0.336	0.066	0.265	0.365	0.398	0.046	0.189	0.587	0.341

Table 4
The final utility values for all of the suppliers derived from the three decision makers

	S_1	S_2	S_3
DM_1	0.496	0.503	0.404
DM_2	0.455	0.477	0.299
DM_3	0.396	0.398	0.279

Table 5
The parameters and the optimal values of the LP model

Supplier index	Utility value	Cost	Capacity	Total demand	Total budget	Order quantity
1	0.447	\$30	650	1200	65 000	550
2	0.457	\$60	650			650
3	0.323	\$35	550			0

In order to find the best order quantities, the TAU which is shown in the following LP model should be maximized:

$$\begin{aligned} \text{Max(TAU)} &= 0.447X_1 + 0.457X_2 + 0.323X_3 \\ \text{Subject to:} \quad & X_1 + X_2 + X_3 = 1200 \\ & 0.01X_1 + 0.02X_2 + 0.04X_3 \leqslant 48 \\ & 50X_1 + 60X_2 + 35X_3 \leqslant 65000 \\ & X_1 \leqslant 650 \\ & X_2 \leqslant 650 \\ & X_3 \leqslant 550 \\ & X_1, X_2, X_3 \geqslant 0 \end{aligned}$$

This LP problem is solved using Lingo 8.0. Table 5 shows the parameters and optimal values of the LP model. After wards to we refer total cost (the seventh of the performance attributes) as *cost*.

5. Discussion

In this section, after solving the LP model the impact of changes in special supplier (S_1)'s cost on quantity of purchasing (LP model optimal answer) is analyzed, where other variables are assumed constant. In order to calculate the impact, S_1's cost is changed to an acceptable range (\$10–\$100) and then the quantity of purchase is monitored to see how it is changing.

Changes in attributes for each supplier will cause change in the utility score of that supplier. This change will follow the utility functions that have been elicited in previous section. For cost attribute, the utility function is shown as

$$a_i + b_i e^{-c_i x} \tag{12}$$

where a_i,b_i, and c_i are scaling constants and differ for each decision maker. The values of these parameters are listed in Table 6. Fig. 2 shows how the cost utility value of supplier 1 is changing when the cost changes in point of view of three DMs (Fig. 3).

As S_1's cost reaches to its minimum level ($10), it became more desirable for DMs. So its utility is maximized (for cost attribute) and LP model proposes to purchase as much as possible from supplier 1 though to S_1's capacity constraint, which is 650. Remaining demand must be purchased from S_2 because our demand is satisfied and there is no need to purchase from S_3. The optimal solution does not change as long as the relation between U_is is $U_1 \geqslant U_2 \geqslant U_3$.

By increasing the S_1s cost, its utility will decrease; so differential between U_1 and U_2 will be reduced also. When cost reaches $30, U_2 will appear higher than U_1, therefore, the optimal solution will be modified and the model would suggest to purchase 650 unit from S_2 and 550 unit from S_1. This circumstance will continue until the cost becomes $47.2. After this point, the relation between U_i s is $U_2 \geqslant U_3 \geqslant U_1$ and the LP model proposes a purchase from S3. By increasing the cost, the order quantity from S3 will also increase, whereas order quantity from S1 is decreased, and the order quantity from S_2 remains constant. When the cost reaches 85.7, the U_1 becomes less than all so the model proposes no purchase from S_1 and all our needs must be procured from S_2 and S_3.

Table 6
The value of utility function for the cost attribute (S1)

	a_i	b_i	c_i
DM_1	−0.097	1.437	0.027
DM_2	−0.049	1.474	0.034
DM_3	−0.008	1.73	0.054

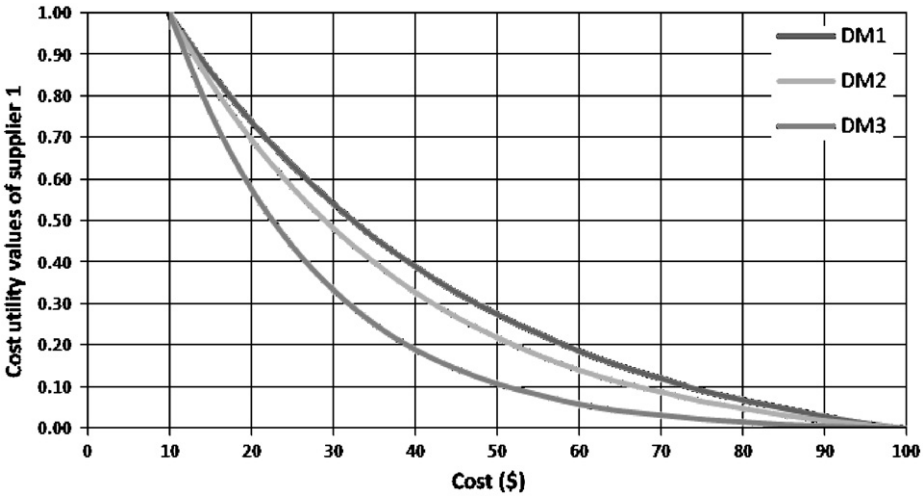

Fig. 2. Cost attribute's utility score for supplier 1.

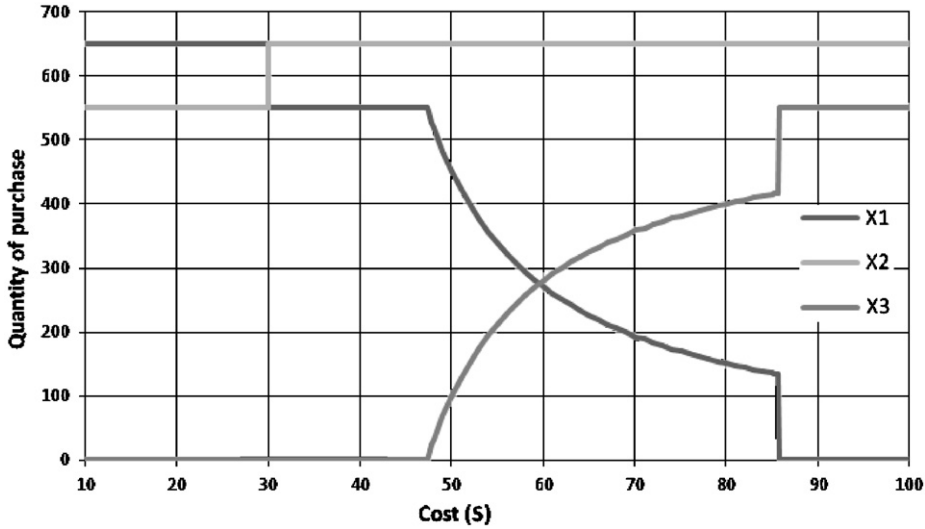

Fig. 3. Impact of changing the S1s cost on the order quantities.

6. Conclusion

Supply chain management has recently received considerable attention in business-management literature. Many companies consider a well-designed and implemented supply chain system as an important tool. Supplier selection is a fundamental aspect of supply chain management, which heavily contributes to the overall supply chain performance.

Supplier selection is a problem with a more complex nature. The problem is a complex multi-criteria decision problem that includes both qualitative and quantitative factors, which are often assessed with imprecise data and human judgments. It appears that MAUT is well suited to deal with such decision problems. This paper presents an effective model using both MAUT and LP for solving this problem. MAUT determines the supplier's utility from the decision makers' viewpoints. The LP model is then used to determine the order quantities to be purchased from each supplier to maximize the quantity of purchase from the most desired suppliers.

The MAUT and LP the model is rather simple to use and meaningful. The systematic framework for supplier evaluation and selection presented in this paper can easily be extended to analyze other managerial decision-making problems. However, the model would be improved and restructured by considering the existing constraints according to the specific needs and the desirable requirements. In addition, supplementary factors such as discount, lot sizing and logistic costs can be considered as a future research.

References

[1] G.W. Dickson, An analysis of vendor selection systems and decisions, J. Purchasing 2 (1966) 5–17.
[2] C.P. Roa, G.E. Kiser, Educational buyer's perception of vendor attributes, J. Purchasing Mater. Manage. 16 (1980) 25–30.

[3] L.M. Ellram, The supplier selection decision in strategic partnerships, J. Purchasing Mater. Manage. 26 (4) (1990) 8–14.
[4] C.L. Stamm, D.Y. Golhar, JIT purchasing attribute classification and literature review, Prod. Plann. Control 4 (3) (1993) 273–282.
[5] C. Weber, L. Ellram, Supplier selection using multi-objective programming: a decision support system approach, Int. J. Phys. Distrib. Logistics Manage. 23 (1992) 3–14.
[6] J. Liu, F. Ding, V. Lall, Using data envelopment analysis to compare suppliers for supplier selection and performance improvement, Int. J. Supply Chain Manage. 5 (3) (2000) 143–150.
[7] L. Forker, D. Mendez, An analytical method for benchmarking best peer suppliers, Int. J. Oper. Prod. Manage. 21 (1–2) (2001) 195–209.
[8] R.F. Saen, A. Memariani, F.H. Lotfi, Determining relative efficiency of slightly non-homogeneous decision making units by data envelopment analysis: a case study in IROST, Appl. Math. Comput. 165 (2) (2005) 313–328.
[9] S. Talluri, R. Narasimhan, Vendor evaluation with performance variability: a max–min approach, Eur. J. Oper. Res. 146 (2003) 543–552.
[10] T.L. Saaty, The Analytic Hierarchy, McGraw-Hill, New York, 1980.
[11] R. Narasimhan, An analytic approach to supplier selection, J. Purchasing Supply Manage. 1 (1983) 27–32.
[12] F.J.B. Partovi, A. Banerjee, Application of analytic hierarchy process in operations management, Int. J. Oper. Prod. Manage. 10 (1989) 5–19.
[13] R. Nydick, R. Hill, Using the analytic hierarchy process to structure the supplier selection procedure, J. Purchasing Mater. Manage. 25 (2) (1992) 31–36.
[14] G. Barbarosoglu, T. Yazgaç, An application of the analytic hierarchy process to the supplier selection problem, Prod. Inventory Manage. J. 38 (1997) 14–21.
[15] S. Yahya, B. Kingsman, Vendor rating for an entrepreneur development program: a case study using the analytic hierarchy process method, J. Oper. Res. Soc. 51 (1999) 916–930.
[16] C. Masella, A. Rangone, A contingent approach to the design of vendor selection systems for different types of cooperative customer/supplier, Int. J. Oper. Prod. Manage. 20 (2000) 70–84.
[17] M. Tam, V. Tummala, An application of the AHP in vendor selection of a telecommunications system, Omega 29 (2001) 171–182.
[18] E. Lee, S. Ha, S. Kim, Supplier selection and management system considering relationships in supply chain management, IEEE Trans. Eng. Manage. (48) (2001) 307–318.
[19] F. Liu, H. Hai, The voting analytic hierarchy process method for selecting supplier, Int. J. Prod. Econ. 97 (3) (2005) 308–317.
[20] J. Sarkis, S. Talluri, A model for strategic supplier selection, J. Supply Chain Manage. 38 (2002) 18–28.
[21] H. Min, International supplier selection: a multi-attribute utility approach, Int. J. Phys. Distrib. Logistics Manage. 24 (5) (1994) 24–33.
[22] A. Pan, Allocation of quantities among suppliers, J. Purchasing Mater. Manage. 25 (1989) 36–39.
[23] S. Ghodsypour, C. O'Brien, A decision support system for supplier selection using an integrated analytic hierarchy process and linear programming, Int. J. Prod. Econ. 56–57 (1998) 199–212.
[24] S. Ghodsypour, C. O'Brien, The total cost of logistics in supplier selection under conditions of multiple sourcing, multiple criteria and capacity constraint, Int. J. Prod. Econ. 73 (2001) 15–27.
[25] J. Current, C. Weber, Application of facility location modelling constructs to vendor selection problems, Eur. J. Oper. Res. 76 (1994) 387–392.
[26] F. Buffa, W. Jackson, A goal programming model for purchase planning, J. Purchasing Mater. Manage. 19 (1983) 27–34.
[27] G. Wang, S. Huang, J. Dismukes, Product-driven supply chain selection using integrated multi-criteria decision-making methodology, Int. J. Prod. Econ. 91 (1) (2004) 1–15.
[28] H. Ding, L. Benyoucef, X. Xiu, A simulation optimization methodology for supplier selection problem, Int. J. Comput. Integrated Manuf. 18 (2–3) (2005) 210–224.
[29] C. Weber, J. Current, A. Desai, An optimization approach to determining the number of vendors to employ, Int. J. Supply Chain Manage. 5 (2) (2000) 90–98.
[30] W. Xia, Z. Wu, Supplier selection with multiple criteria in volume discount environments, Omega 35 (5) (2006) 494–504.
[31] R. Keeney, H. Raiffa, Decisions with Multiple Objectives: Preferences and Value Tradeoffs, Wiley, New York, 1976.

[32] R. Stewart, S. Mohamed, IT/IS project selection using multi-criteria utility theory, Logistics Inf. Manage. 15 (4) (2002) 254–270.
[33] S. Zionts, Some thoughts on research in multiple criteria decision making, Comput. Oper. Res. 19 (7) (1992) 567–570.
[34] N. Aissaoui, M. Haouaria, H. Elkafi, Supplier selection and order lot sizing modeling: a review, Comput. Oper. Res. 34 (12) (2007) 3516–3540.
[35] S.E. Bodily, A delegation process for combining individual utility functions, Manage. Sci. 25 (10) (1979) 1035–1041.
[36] S.H. Huang, H. Keskar, Comprehensive and configurable metrics for supplier selection, Int. J. Prod. Econ. 105 (2) (2007) 510–523.
[37] L. de Boer, E. Labro, P. Morlacchi, A review of methods supporting supplier selection, Eur. J. Purchasing Supply Manage. 7 (2001) 75–89.